UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	Keyframes define important positions in an animation.	Keyframes mark major changes.
A2	FALSE	Higher resolution always means a smaller file.	Higher resolution usually increases file size.
A3	FALSE	A plotter is an input device.	A plotter produces output.
A4	TRUE	A storyboard is a visual plan of a multimedia sequence.	Storyboards show the planned flow.
A5	FALSE	Anti-aliasing makes the edges of text look more jagged.	Anti-aliasing smooths jagged edges.
A6	FALSE	3D animation does not use modelling.	3D animation relies on models.
A7	TRUE	A bitmap image is made up of pixels.	Bitmap images use a pixel grid.
A8	TRUE	Authoring tools help developers organise and edit multimedia elements.	Authoring tools assemble media.
A9	TRUE	Interactivity gives users control over content and flow.	Interactivity is a defining feature of many multimedia products.
A10	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A11	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A12	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A13	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.
A14	TRUE	A URL identifies the address of a web resource.	A URL is a resource locator.
A15	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	F	Bitmap	image made of pixels
B2	D	Storyboard	visual sequence plan
B3	L	Sampling	converting sound into digital values
B4	K	Keyframe	important frame in an animation
B5	B	Vector	image made of mathematical objects
B6	G	Scanner	input device that digitises documents
B7	N	Tweening	generating in-between frames
B8	O	Virtual reality	simulated environment
B9	I	Serif	font with small strokes at character ends
B10	C	MIDI	standard for musical notes and instructions
B11	M	Card-based tool	authoring tool organised like pages or cards
B12	J	Browser	software that displays web pages
B13	A	Multimedia	integration of text, graphics, audio, video and animation
B14	H	Cel animation	hand-drawn frame-by-frame animation
B15	E	Plotter	output device for large drawings

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	D	The user can control content and flow
C2	A	Play while downloading
C3	C	Replace a colour background
C4	C	Important frames in an animation
C5	A	WAV
C6	A	Structure web content
C7	B	A video compression family
C8	D	Multimedia with hyperlinks
C9	D	Messages with a beginning and end
C10	B	Drawing input
C11	A	Frames per second
C12	C	A low-fidelity layout sketch
C13	D	Draws alternate lines of a frame
C14	D	Smooth jagged edges
C15	A	Printed output
C16	C	Improves accuracy
C17	D	Pixels
C18	A	Display web pages
C19	D	Non-linear
C20	A	Space between individual character pairs
C21	B	Merging of media and computing
C22	B	Permission to use a work
C23	A	Temporary working memory
C24	D	An early model of a product
C25	B	Scanner
C26	A	Drawing each frame by hand
C27	A	A cinema presentation
C28	D	Analog sound to digital values
C29	A	Loudness
C30	C	Number of pixels in an image

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PART D - WORKED ANSWERS

QUESTION 1

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

QUESTION 2

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

QUESTION 3

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(2 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(4 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(4 marks)

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QUESTION 4

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(5 marks)

END OF MARKING SCHEME

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